

DDA 5002

Sample Midterm Exam

Fall 2025

Name (please print): _____

This is a closed-book, closed-notes, and closed-calculator exam. No formula sheets, or other materials are allowed. If you need additional scratch paper, you should ask the proctor.

Where possible, show your work, and explain your thinking (for example, label constraints and define decision variables). Every attempt will be made to award partial credit for incorrect or incomplete answers, but evidence of your work and your thought process must be apparent. All answers and supporting work must be your own.

You have two hours to complete this exam. Good luck!

1. (20 pts) Consider each of the following situations independently. Formulate one or more linear programming constraints to capture the stated condition. Introduce, and clearly define, additional decision variables if necessary.

(a) (10 pts) Let x_1, x_2 be decision variables, and let $a_{11}, a_{21}, a_{12}, a_{22}$, and b be scalar parameters:

$$|x_1 - c| + \max\{a_{11}x_1 + a_{12}x_2, a_{21}x_1 - a_{22}x_2\} \leq b$$
$$x_1, x_2 \geq 0$$

Rewrite these constraints using (a reasonable number of) linear programming constraints.

(b) (10 pts) Consider a regression problem of fitting a function $y = ax + b$ to data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. We wish to estimate a and b to give us the best fit. Use the one-norm; recall that for vectors p, q of length m , this is defined as:

$$\|p - q\|_1 = \sum_{i=1}^m |p_i - q_i|$$

In other words, minimize the sum of the absolute values of the difference between the data and our predicted values for $y_1 \dots y_n$. Formulate this as a non-linear program, as in class. Then transform it into an appropriate linear program using one of the two methods that were discussed.

2. (20 pts) Sweet-Tooth Inc. makes candy bars called Candy1 and Candy2. Both bars are manufactured using a blend of sugar, nuts, and chocolate. The pictured table specifies the allowable composition of each bar, the range of sales we can make, and the revenue per bar. “Max” denotes the largest amount of an ingredient that can go into a bar, while “Min” denotes the minimum. Demand and revenue are also given, such that production of each bar must satisfy the bounds on their respective demand.

Product Type	Sugar [ounces/bar]		Nuts [ounces/bar]		Chocolate [ounces/bar]		Demand [bars]		Revenue [\$ /bar]
	Min	Max	Min	Max	Min	Max	Min	Max	
Candy1	l_{s1}	u_{s1}	l_{n1}	u_{n1}	l_{c1}	u_{c1}	d_1	D_1	r_1
Candy2	l_{s2}	u_{s2}	l_{n2}	u_{n2}	l_{c2}	u_{c2}	d_2	D_2	r_2

We can purchase up to b_s , b_n , and b_c ounces of sugar, nuts and chocolate, respectively. They have a cost of k_s , k_n , and k_c , respectively. Lastly, a Candy1 bar must weigh exactly f_1 ounces, while a Candy2 bar must weigh exactly f_2 ounces. Formulate a data-independent linear program to maximize profit while obeying the bounds on ingredients, as well as demand for each product.

Note: You will find it easier to write composition and weight constraints as an average over all bars produced. For example, the total weight of all type-1 bars should be f_1 *[number of type 1 bars produced], instead of saying that each type-1 bar must weigh exactly f_1 . The latter type of constraint will lead to a non-linear program, which we will see later in the class.

3. (20 pts) Consider the following linear program:

$$\max \quad 4x_1 + 5x_2 + x_3 \quad (1a)$$

$$\text{s.t.} \quad x_1 + x_2 + x_3 - x_4 = 1 \quad (1b)$$

$$2x_1 + 3x_2 + x_3 + x_5 = 8 \quad (1c)$$

$$4x_1 + 4x_2 - 3x_3 + x_6 = 4 \quad (1d)$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0. \quad (1e)$$

- (a) (8 pts) Suppose you are solving the phase-I problem, and you are only allowed to add one auxiliary variable. Write down the phase-I LP formulation. Perform the simplex method to solve this phase-I problem. Report the starting basic feasible solution and the corresponding basis for the phase-II problem.
- (b) (6 pts) Suppose at an iteration of the simplex process for the phase-II problem, the basic variables are $\{x_1, x_3, x_4\}$. What is the solution corresponding to this basis? Is this solution optimal? If so, explain why it is an optimal solution. If not, what is the search direction d ? How far can you move along that search direction?
- (c) (6 pts) Following part (b), if you change the objective function (1a) to

$$\max \quad 4x_1 + 6x_2 + x_3,$$

is the current solution corresponding to the basic variables $\{x_1, x_3, x_4\}$ optimal? If so, explain why it is an optimal solution. If not, what is the search direction d ? How far can you move along that search direction?

4. (20 pts) We are considering building some combination of plants of type 1 and 2 in regions a , b , and c . Let x_{ij} be a binary decision variable that takes value 1 if we build a plant of type i in region j and takes value 0 otherwise, where $i \in \{1, 2\}$ and $j \in \{a, b, c\}$. Consider each of the following situations independently. Formulate one or more *linear* integer constraints (and objective in part (g)) that models the stated condition. Introduce, and clearly define, additional decision variables if necessary. Please use a data-specific formulation.

- (a) (2 pts) At least one plant must be built in each region.
- (b) (2 pts) At least one plant of type 1 must be built.
- (c) (3 pts) If region a contains a plant of type 1 then region c must also contain a plant of type 1.
- (d) (3 pts) Region a cannot contain a plant of type 1 and of type 2.
- (e) (3 pts) Region b can contain a plant of type 2 only if: a plant of type 1 is built in region a and a plant of type 2 is built in region c .
- (f) (4 pts) Region c contains a plant of type 2 if and only if region a and region b both contain a plant of type 2; i.e., $x_{2c} = x_{2a} \cdot x_{2b}$. This is a *nonlinear* constraint. You are to reformulate it using one or more *linear* constraints.
- (g) (3 pts) The objective is to maximize the number of regions that contain plants of both types.

5. (20 pts) Suppose there are three types of raw material, from which manufacture three types of products. The amount of raw material consumed per unit (in kg) and profit per unit (in \$), for each type of the products, are shown in the following table.

Product	Raw Material Consumption per Unit [kg]			Profit per Unit [\$]
	1	2	3	
1	1	2	0	2
2	3	1	1	4
3	0	0	4	1

Given that you have 8 kg of raw material 1, 6 kg of raw material 2, and 6 kg of raw material 3, you can write down the following data-specific linear program to model the problem.

$$z^P = \max \quad a_1 x_1 + 4x_2 + x_3 \quad (2a)$$

$$\text{s.t.} \quad x_1 + a_2 x_2 \leq 8 \quad (2b)$$

$$2x_1 + x_2 \leq b \quad (2c)$$

$$x_2 + a_3 x_3 \leq 6 \quad (2d)$$

$$x_1, x_2, x_3 \geq 0. \quad (2e)$$

- (a) (3 pts) Determine the values of a_1 , a_2 , a_3 , and b .
- (b) (3 pts) Derive the dual problem of the formulation above, i.e., model (2).
- (c) (3 pts) Write the complementary slackness conditions of this model (both primal and dual parts).
- (d) (3 pts) After solving this problem in Python, you have obtained an optimal solution as:

$$\begin{bmatrix} x_1^* \\ x_2^* \\ x_3^* \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, BV = \{x_1, x_2, x_3\}.$$

Use the complementary slackness conditions to derive an optimal solution to the dual.

- (e) (3 pts) If you can purchase 1 kg of raw material 1 with \$1, will you make the purchase? Explain.
- (f) (5 pts) If you can increase the value of b (the amount of raw material 2 available), at what value the current basis will no longer remain optimal; i.e., what is the largest b so that x_1 , x_2 and x_3 are still basic variables? Explain.